Ontario Tech University

# Big Data System Design (MBAI 5110G)

Winter-2026

## Week 7: Big Data Pipeline


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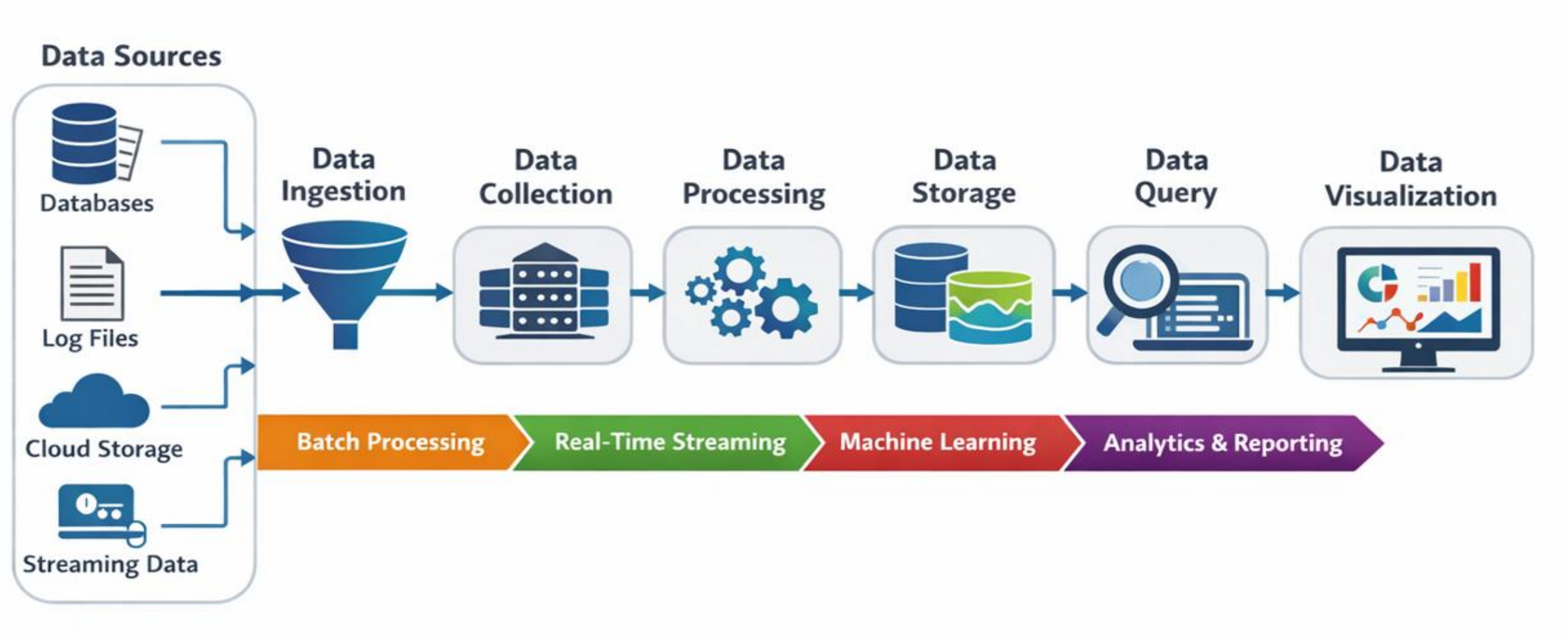
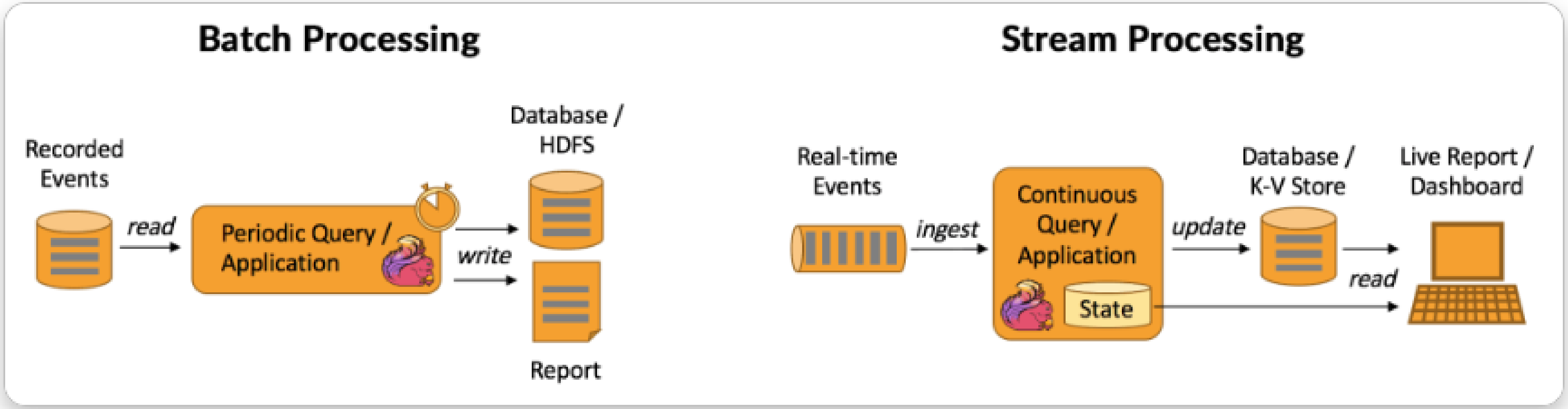
# Outline

- Big data pipeline concept
- Batch vs Stream Processing
- Spark Data Pipeline
- Spark vs Traditional ETL
- Spark Declarative Pipelines
- Steps to build Spark Data Pipeline
- Big-Data Pipeline Examples

# Big Data Pipeline Concept



# Batch vs Stream Processing



# What is a Spark Data Pipeline?

- A **Spark data pipeline** is a workflow that:
  - Ingests raw data
  - Processes and transforms it
  - Stores it for access
  - Makes it available for analysis & reporting
- It distributes processing across multiple servers for large-scale data.

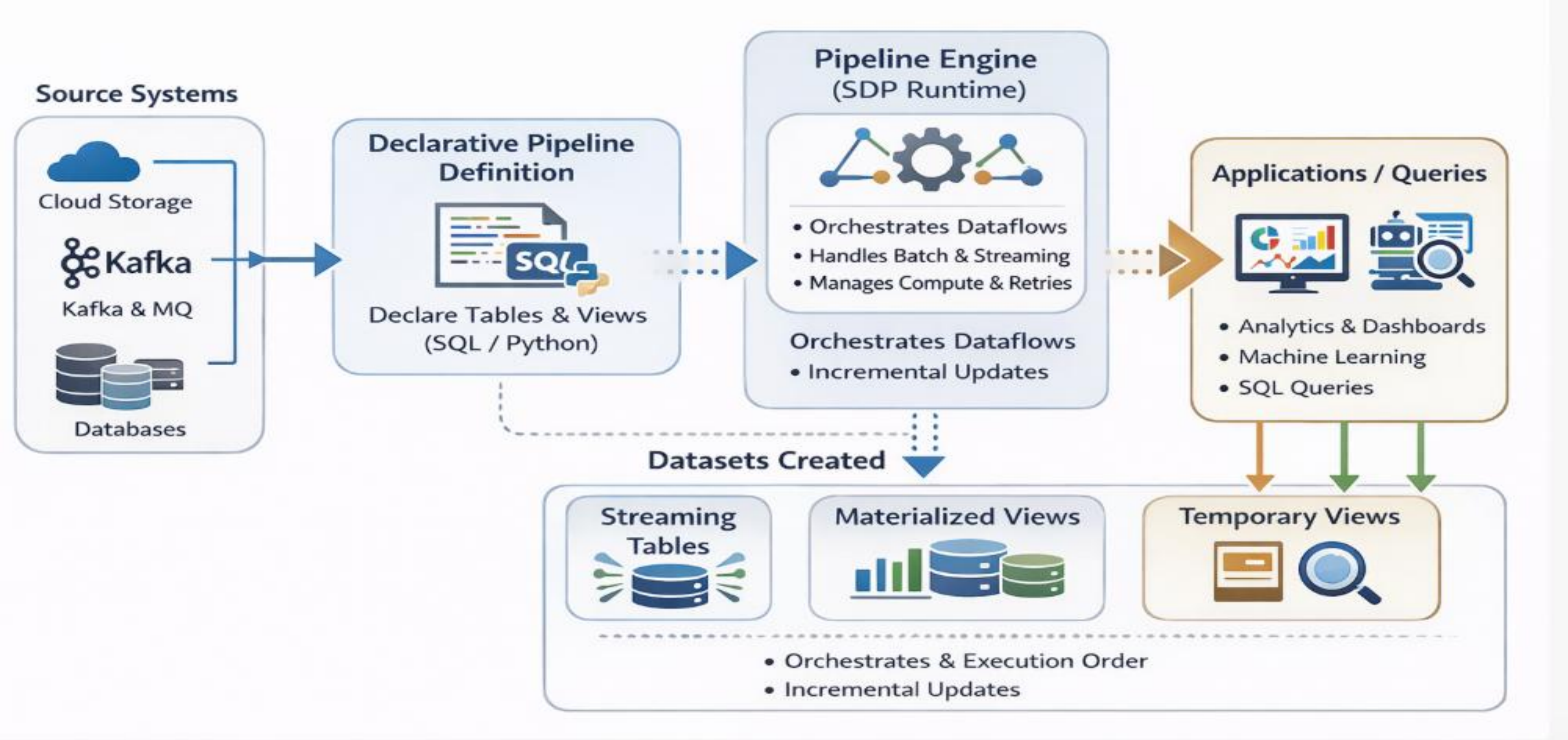
- **Core Elements of Spark Data Pipeline**:
  - Data Sources :
    - Databases, Kafka, Logs, Cloud Storage
  - Data Processing:
    - RDDs and DataFrames
  - Data Storage:
    - Data Lake, Warehouse, HDFS, S3
- **These components form the backbone of scalable analytics**

# Spark Pipeline vs Traditional ETL

- **Traditional ETL:**
  - Batch-only processing
  - Single-server limitations
  - Scheduled updates
- **Spark Pipeline:**
  - Batch + Real-time streaming
  - Distributed cluster processing
  - Supports ML, SQL, and streaming in one system

# Spark Declarative Pipelines

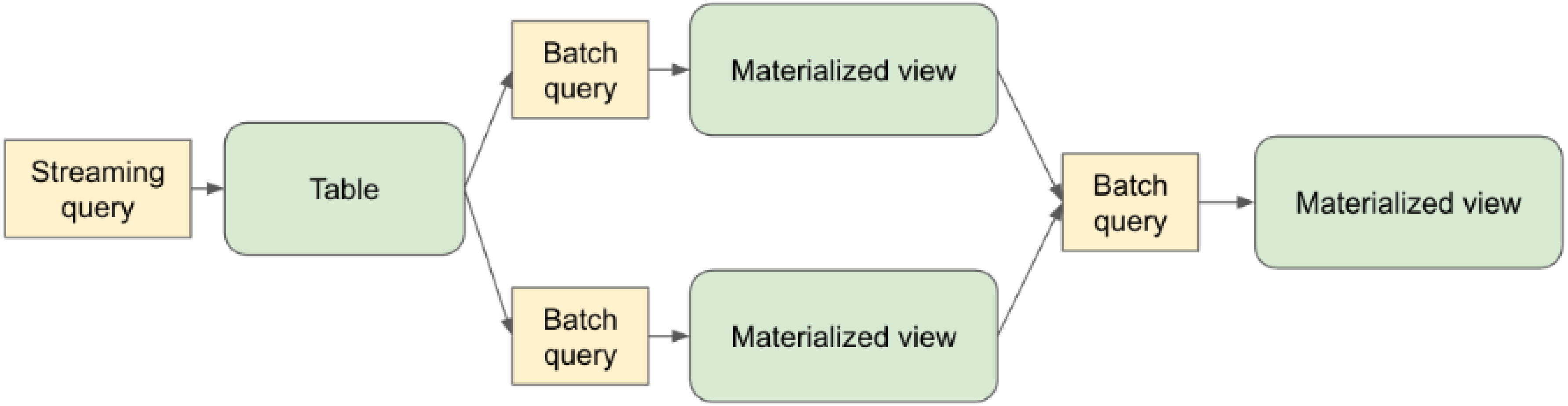
- Spark Declarative Pipelines (SDP)[1] is a declarative framework for building reliable, maintainable, and testable data pipelines on Spark




[1] https://spark.apache.org/docs/latest/declarative-pipelines-programming-guide.html#what-is-spark-declarative-pipelines-sdp

# Spark Declarative Pipelines (Cont.)

- **SDP** supports:
  - **Data ingestion** from cloud storage:
    - Amazon S3, Azure ADLS Gen2, Google Cloud Storage
  - Data ingestion from message buses:
    - Apache Kafka, Amazon Kinesis, Google Pub/Sub, Azure EventHub
  - Incremental batch and streaming transformations (batch and streaming )
- The **key advantage of SDP** is its **declarative approach:**
  - You define what tables should exist and what their contents should be
  - SDP handles the orchestration, compute management, and error handling automatically

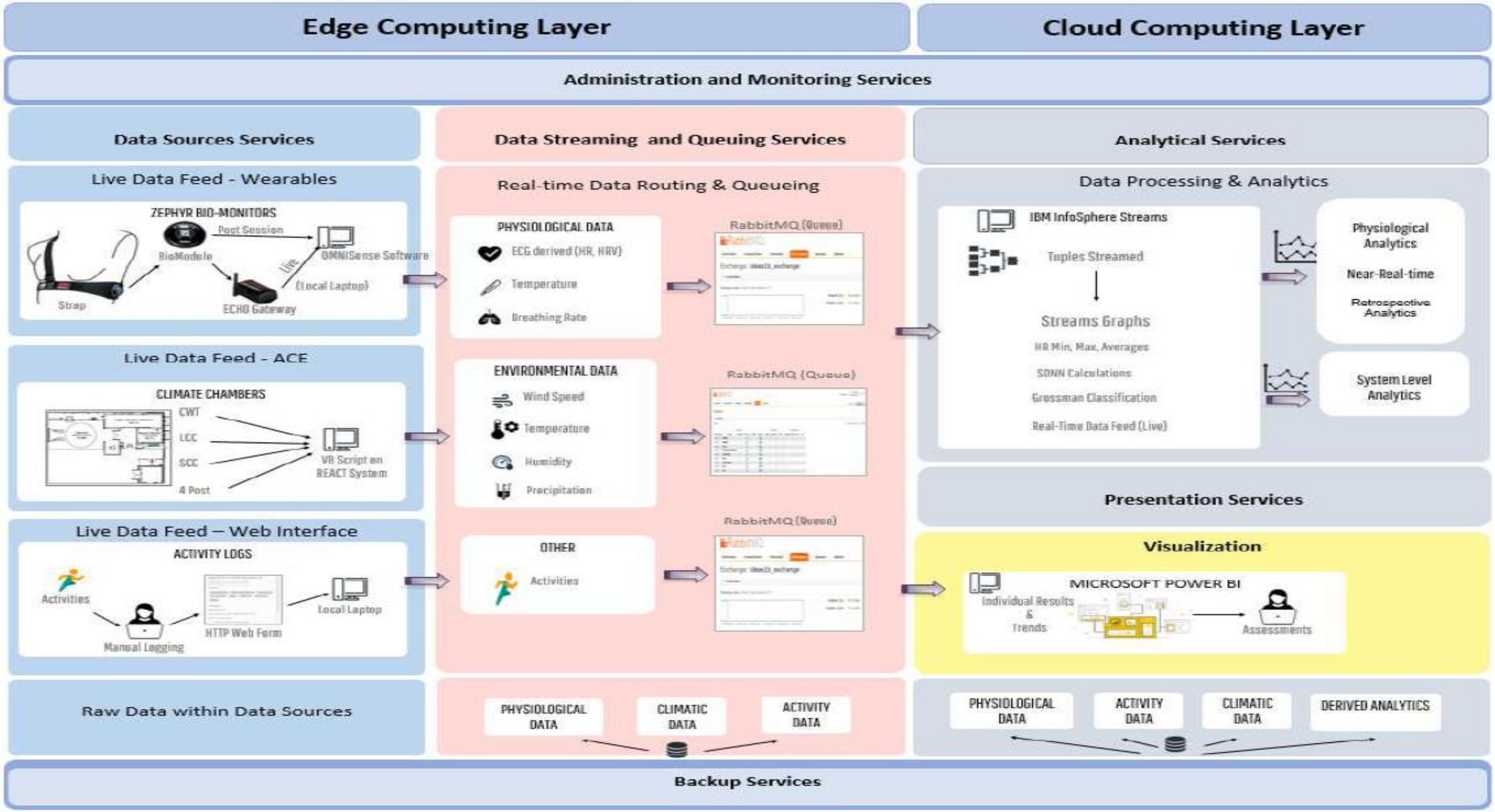


# Steps to Build a Spark Data Pipeline

- **Step 1** — Data Ingestion (JDBC, Kafka, S3, etc.)
- **Step 2** — Data Collection Layer (RDD/DataFrame)
- **Step 3** — Data Processing (Batch or Streaming)
- **Step 4** — Data Storage (Lake/Warehouse)
- **Step 5** — Data Query (Spark SQL)
- **Step 6** — Data Visualization (Dashboards & Reports)

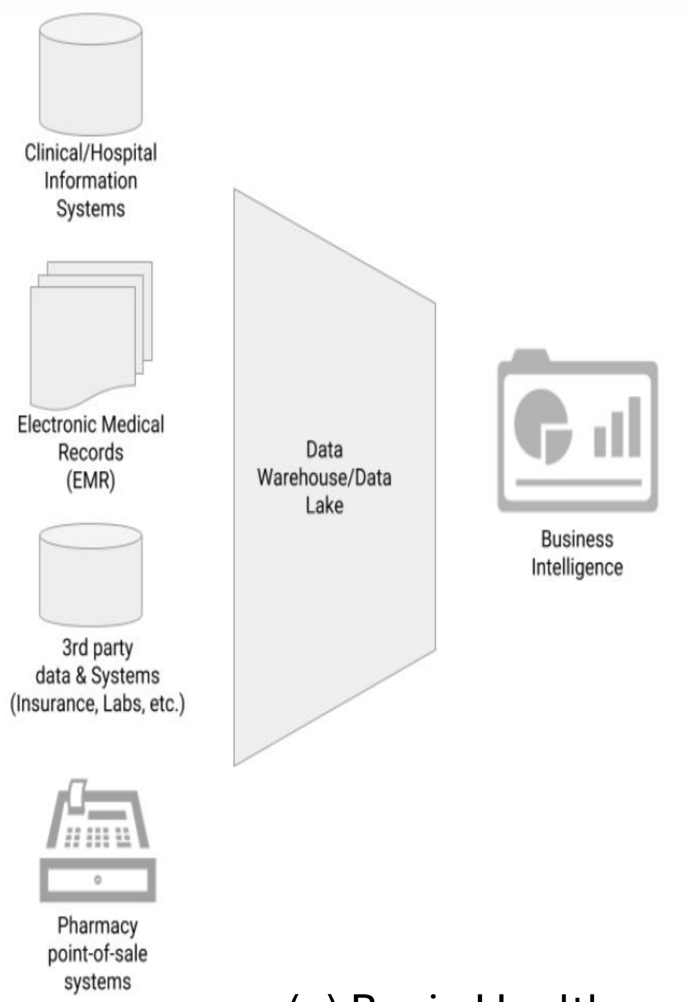
# Some Examples of Big Data Pipelines

# Big Data Pipeline 1: Case Study in Extreme Climatic Human Studies[3]




[3]Inibhunu, Catherine, Jennifer Yeung, Aaron Gates, Bradley Chicoine, and Carolyn McGregor. "A Decoupled Data Pipeline and its Reliability Assessment: Case Study in Extreme Climatic Humans Studies." In 2021 IEEE International Conference on Big Data (Big Data), pp. 3075-3084. IEEE, 2021

# Big Data Pipeline 2: Big Data for Health Care with Data Science Pipelines[4]




(a) Basic Healthcare


(b) Modern Healthcare

[4]https://www.cloudgeometry.com/blog/beyond-big-data-for-health-care-with-data-science-pipelines

# Next Session: Big Data Analytics

# Q/A

Thank you!